LAB 1

Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #56 : Media, Events & Community
Peer Skill Exchange & Mentorship Network

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FUNCTIONAL AND NON FUNCTIONAL REQUIREMENTS

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	Maintaining a time-credit metric for each member, adding 1 hour upon verified completion of a teaching session.	High	Pass: 1 time credit is transferred from the learner to the mentor after both confirm session completion. Fail: A member with 0 available credits cannot book a session requiring a credit.	Core functionality of the time-banking skill exchange platform.
FR-002	Functional	The system shall allow a Learner to search for and select a mentor based on the skill they want to learn.	High	Pass: When a learner searches for a skill the system displays at least one matching mentor when available and the learner can select a mentor. Fail: The system allows selection of a mentor who does not offer the requested skill.	Enables members to find suitable mentors and initiate skill exchanges.
FR-003	Functional	The system shall allow a Learner and mentor to schedule a skill-learning session by selecting a mutually available date and time.	High	Pass: A session is created only when the selected time is available for both members, and the confirmed session appears in their schedules. Fail: The system creates a session during a time marked unavailable.	Scheduling is necessary for members to arrange and conduct skill-exchange sessions.

FR-004	Functional	The system shall allow the Learner and mentor to confirm completion of a scheduled teaching session before transferring the corresponding time credit.	High	Pass: Time credit is transferred only after the required session completion confirmation is recorded. Fail: Credit is transferred when the session has not been confirmed as completed.	Ensures that time credits are exchanged only for verified teaching sessions.
FR-005	Functional	The system shall allow both the Learner and mentor to submit a rating and feedback after a verified skill-exchange session.	Medium	Pass: After a session is verified as completed, each participant can submit one rating and feedback. Fail: A participant can submit feedback before the session is completed or submit multiple ratings for the same session.	Supports mutual feedback and rating validation, which is explicitly part of the platform.
NFR-001	Nonfunc Performanc e & Security	The session scheduling calendar must be in sync with Google Calendar and Outlook via iCal export feeds.	High	Pass: A scheduled session is correctly represented in the exported iCal feed and can be imported into Google Calendar or Outlook without losing the session date, time, or details. Fail: The exported event contains missing or incorrect scheduling information.	Allows members to integrate skill-exchange sessions with commonly used calendars.
NFR-002	Nonfunction al — Security	The system shall protect member account and session information so that only authorized members can access their own personal scheduling and time-credit data.	High	Pass: An authenticated member can access their own time-credit and scheduling information, while an attempt to access another member's private information is denied. Fail: An unauthorized member can view or modify another member's private data.	Protects sensitive member and session information and prevents unauthorized access.

ACTORS INVOLVED

Actor 1 Learner Member
The member who wants to learn a skill from another member.
what they do : Search for mentor, select a mentor, schedule a session, use time-credits, confirm session completion, give feedback/rating

Actor 2 Mentor teaches a skill to another member.
What they do : provide a skill, Participate in scheduling, Conduct teaching sessions, tell session completion, receive time credits, Give feedback/rating

Actor 3 Calendar Service
We know the system must synchronize with Google Calendar Outlook via iCal export feeds. We can represent the external calendar systems with a general actor. This is an external system that interacts with our Peer Skill Exchange system.

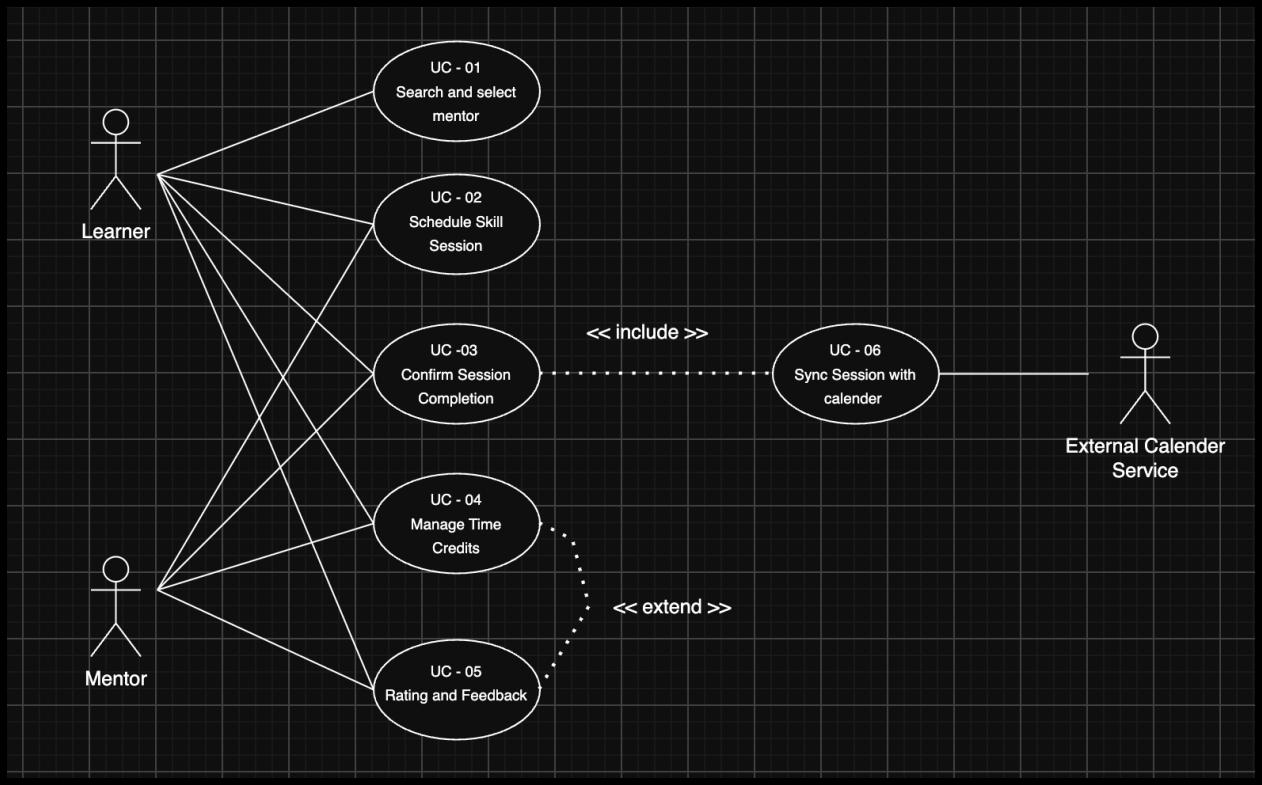
Use Case ID	Use Case Name	Description	Actor(s)
UC-01	Search and Select Skill Mentor	Allows a Learner Member to search for a required skill and select a suitable Skill Mentor.	Learner Member
UC-02	Schedule Skill Session	Allows the Learner Member and Skill Mentor to schedule a session at a mutually available date and time.	Learner Member, Skill Mentor
UC-03	Confirm Session Completion	Allows the Learner Member and Skill Mentor to confirm that a scheduled teaching session has been completed.	Learner Member, Skill Mentor
UC-04	Manage Time Credits	Maintains and updates member time-credit balances following verified completion of teaching sessions.	Learner Member, Skill Mentor
UC-05	Submit Rating and Feedback	Allows the Learner Member and Skill Mentor to submit feedback and ratings after a verified session.	Learner Member, Skill Mentor
UC-06	Sync Session with Calendar	Synchronizes scheduled skill-exchange sessions with external calendar services using iCal export feeds.	ExternCalendar Service

UML RELATIONSHIPS

UC-02 Schedule Skill Session «include» UC-06 Sync Session with Calendar, because calendar synchronization is performed as part of scheduling a session.

UC-05 Submit Rating and Feedback «extend» UC-03 Confirm Session Completion, because providing feedback is an optional behavior after a session has been completed.

UML DIAGRAM



Use-Case Flow Document

1. Use Case Information

Use Case ID: UC-02 Use Case Name: Schedule Skill Session Primary Actors: Learner Member, Skill Mentor Supporting Actor: External Calendar Service

Description: This use case allows a Learner Member and a Skill Mentor to agree on a mutually available date and time for a skill-learning session. Once confirmed, the session is synchronized with an external calendar service for both participants.

2. Preconditions

- The Learner Member has selected a suitable Skill Mentor for the desired skill.
- Both the Learner Member and Skill Mentor have valid, active accounts on the platform.
- The Skill Mentor has provided their availability for scheduling.

3. Postconditions

- The skill-learning session is successfully scheduled with an agreed date and time.
- The session details are recorded for both the Learner Member and Skill Mentor.
- The session is synchronized with the external calendar service.
- Both participants receive confirmation of the scheduled session.

4. Main Success Scenario

1. The Learner Member initiates scheduling for a session with the selected Skill Mentor.
2. The system retrieves and displays the Skill Mentor's available time slots.
3. The Learner Member selects a preferred date and time from the displayed options.
4. The system verifies that the selected time slot is still available.
5. The system requests confirmation from the Skill Mentor for the selected time slot.
6. The Skill Mentor reviews and confirms the proposed session time.
7. The system records the session details for both participants.
8. The system synchronizes the scheduled session with the external calendar service via an iCal export feed.
9. The system displays a confirmation message to both the Learner Member and Skill Mentor.

5. Alternate Flow

4a. Selected time slot is unavailable.

1. The system informs the Learner Member that the selected time slot is no longer available.
2. The system displays the updated list of available time slots.
3. The Learner Member selects an alternative time slot.
4. The flow resumes at step 4 of the Main Success Scenario.

If the Learner Member does not select an alternative time slot, the use case terminates without scheduling a session.